Biology_2024_PYQ

Group-A (Very Short Answer Type Question)

Answer any ten of the following: [1 x 10 = 10]

1. The portion of the growth curve where rapid growth of bacteria is observed is known as ___.
2. The human eye can focus objects at different distances by adjusting the focal length of the eye lens. This is due to ___.
3. What is meant by the term osmoregulation?
4. Who is known as father of genetics?
5. What is meant by power of accommodation of the eye?
6. Enzyme increases the rate of reaction by lowering the activation energy. Is this statement true or false?
7. What is cistron?
8. State two economically important uses of heterotrophic bacteria.
9. What is the main function of kinase?
10. The growth of bacterial population follows a geometric progression. a) True b) False
11. Are viruses living or non-living?
12. Fluid Thioglycolate medium is used for the cultivation of which type of organism?

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Group-B (Short Answer Type Question)

Answer any three of the following: [5 x 3 = 15]

2. How are co-factors different from prosthetic groups? [5]
3. Give characteristics of genetic code. [5]
4. Differentiate between DNA/RNA. [5]
5. Explain Krebs cycle. Draw suitable flowchart for explanation. [5]

6. The sequence of the coding strand of DNA in a transcription unit is mentioned below: 3' AATGCAGCTATTAGG 5'.

Write the sequence for:

1. Its complementary strand.

2. Its mRNA. [5]

Group-C (Long Answer Type Question)

Answer any three of the following: [15 x 3 = 45]

7. Illustrate the two models by which an enzyme holds the substrate. [15]

8. Explain steps of Glycolysis in details. [15]

9. Give functions of Proteins as receptors and structural elements. [15]

10. Describe the characteristics of the individuals with the following chromosomal abnormalities:

Trisomy at chromosome 21;

XXY;

XO. [15]

11. A tall plant with red flowers (dominant) is crossed with a dwarf plant with white flowers (recessive). Work out a dihybrid cross and state the dihybrid ratio. What will be the effect on the dihybrid ratio if the two genes are interacting with each other? [15]

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